

DUNHUA, China

WMO: 541860

Lat: 43.3676N

Lon: 128.2110E

Elev: 525

StdP: 95.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-23.9	-30.6	0.2	-24.4	-28.7	0.3	-22.4	8.2	-14.0	7.4	-12.8	1.1	250	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	29.6	20.3	28.1	19.8	26.7	19.2	22.7	27.1	21.8	25.7	21.0	24.6	2.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	17.0	25.2	20.5	16.2	24.3	19.7	15.3	23.5	70.2	27.3	66.7	25.8	63.5	24.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.7	5.8	-28.1	32.1	2.7	1.6	-30.1	33.3	-31.7	34.2	-33.2	35.1	-35.2	36.3		
(5)			-28.3	23.8	2.6	1.0	-30.2	24.5	-31.7	25.1	-33.1	25.7	-35.0	26.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-15.1	-11.0	-3.3	5.9	12.8	18.1	21.2	20.1	14.2	6.1	-4.1	-12.6	(6)	
(7)		DBStd	13.44	4.38	5.35	5.60	4.97	3.96	3.22	2.69	2.80	3.42	4.49	5.99	4.99	(7)	
(8)		HDD10.0	3202	778	588	412	142	17	1	0	0	6	135	422	701	(8)	
(9)		HDD18.3	5281	1037	822	669	374	177	41	7	14	130	379	672	959	(9)	
(10)		CDD10.0	1175	0	0	1	19	103	245	348	314	131	14	0	0	(10)	
(11)		CDD18.3	212	0	0	0	1	5	36	97	69	5	0	0	0	(11)	
(12)		CDH23.3	1500	0	0	0	10	98	332	635	383	41	1	0	0	(12)	
(13)		CDH26.7	319	0	0	0	1	18	76	152	71	2	0	0	0	(13)	
(14)	Wind	WSAvg	2.3	2.2	2.4	2.9	3.0	2.7	2.2	2.0	1.9	1.9	2.3	2.4	2.2	(14)	
(15)	Precipitation	PrecAvg	634	4	6	10	33	60	103	156	152	61	28	15	6	(15)	
(16)		PrecMax	914	14	23	24	72	113	209	284	325	152	60	36	38	(16)	
(17)		PrecMin	359	0	1	0	9	10	21	37	46	13	2	3	1	(17)	
(18)		PrecStd	123	3	6	6	16	28	52	56	72	38	18	9	7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	5.8	15.0	25.0	28.9	31.0	32.1	30.6	26.5	22.1	13.5	3.7	(19)	
MCWB			-3.4	0.5	6.7	12.1	15.4	18.1	22.0	22.3	18.1	12.7	7.2	0.9	(20)		
2%		DB	-3.3	3.0	10.6	20.9	26.2	28.8	29.8	28.5	24.4	19.0	10.1	0.6	(21)		
		MCWB	-5.4	-0.9	3.8	10.2	13.9	17.9	21.5	21.5	16.7	10.9	5.1	-1.8	(22)		
5%		DB	-5.4	0.8	7.9	18.0	23.7	27.0	28.3	26.9	22.7	16.3	7.4	-1.8	(23)		
		MCWB	-7.3	-2.3	2.2	8.7	13.2	17.4	21.0	20.6	15.5	9.6	3.5	-3.9	(24)		
10%		DB	-7.6	-1.5	5.4	15.2	21.1	25.1	26.8	25.4	21.1	14.0	5.0	-4.3	(25)		
		MCWB	-9.2	-4.2	0.7	7.1	12.1	17.1	20.5	20.0	14.6	8.4	1.8	-6.0	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.9	1.2	7.3	14.3	17.3	20.7	24.2	23.6	20.0	13.7	7.8	1.3	(27)	
MCDWB			-0.9	4.2	14.2	21.9	25.4	26.5	29.3	28.7	23.6	20.3	12.1	3.3	(28)		
2%		WB	-5.3	-0.5	4.5	11.0	15.4	19.7	23.1	22.6	18.4	12.0	5.6	-1.5	(29)		
		MCDWB	-3.4	2.6	9.6	20.5	22.8	25.2	27.7	26.9	22.1	17.5	9.4	0.5	(30)		
5%		WB	-7.2	-2.2	2.6	9.1	14.1	18.9	22.2	21.8	17.1	10.6	3.9	-3.9	(31)		
		MCDWB	-5.4	0.9	7.2	16.8	21.3	24.6	26.4	25.5	20.9	15.0	7.1	-2.0	(32)		
10%		WB	-9.2	-4.2	1.0	7.5	12.9	18.0	21.3	21.0	15.8	9.2	1.9	-5.9	(33)		
		MCDWB	-7.6	-1.6	5.0	14.3	20.0	23.3	25.3	24.1	19.7	13.3	4.8	-4.2	(34)		
(35)	Mean Daily Temperature Range			MDBR	11.0	11.7	10.5	11.4	11.5	10.4	8.6	8.8	11.9	12.0	9.7	9.9	(35)
5% DB				MCDBR	13.3	13.6	13.3	16.4	16.5	14.0	11.5	10.8	14.1	16.0	12.9	12.0	(36)
				MCWBR	11.4	10.3	7.8	8.1	6.6	5.2	4.4	4.5	6.7	8.7	8.5	10.0	(37)
5% WB				MCDBR	13.2	12.9	12.4	14.9	13.8	11.1	9.1	8.9	10.5	14.0	11.9	11.7	(38)
				MCWBR	11.4	9.9	7.8	7.8	6.2	4.9	4.1	4.3	5.6	8.3	8.4	10.0	(39)
(40)		Clear-Sky Solar Irradiance	taub		0.333	0.380	0.457	0.542	0.584	0.589	0.553	0.491	0.438	0.464	0.397	0.347	(40)
(41)	taud		2.038	1.935	1.823	1.724	1.689	1.756	1.896	2.056	2.113	1.919	1.986	2.018	(41)		
(42)	Ebn at Noon		753	777	762	731	715	714	735	770	780	677	663	687	(42)		
(43)	Edh at Noon		109	145	185	221	236	221	191	157	137	147	116	102	(43)		
(44)	All-Sky Solar Radiation	RadAvg		2.34	3.37	4.40	4.70	5.20	5.37	4.84	4.68	4.32	3.11	2.08	1.84	(44)	
(45)		RadStd		0.17	0.18	0.28	0.33	0.45	0.55	0.50	0.43	0.34	0.25	0.20	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-0.96	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page